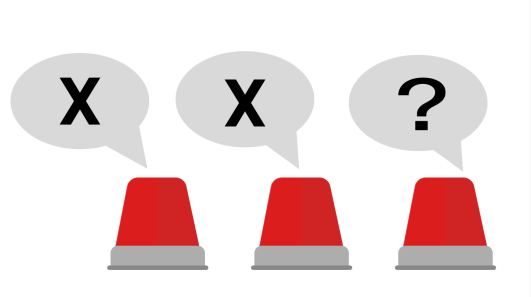
The Discrepant Sensor

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes



Marina has just been promoted to lead engineer in the Quality Control department of a renowned automotive components manufacturer. After years of dedication to the company and overcoming technical challenges, she now faces her first major mission: solving a critical problem in the high-precision sensor production line.

The factory received an urgent alert from an important client: vehicles equipped with sensors from the latest shipment are showing unexpected behaviors. "The cars are displaying inconsistent readings, and this is affecting the autonomous braking system," reported the concerned client, worried about driver safety.

Marina was called to the central control room, where engineers monitor the triple calibration process — the company's pride. This system uses three independent sensors to measure the same variable, ensuring maximum reliability through redundancy. Under normal conditions, all should present identical or very similar readings. When there is a significant discrepancy, it means one of the sensors is faulty.

The diagnostic terminal displays three integer values K_1, K_2, and K_3. Time is crucial — every minute lost could mean more defective sensors being dispatched, potentially putting lives at risk. Marina knows that in each set of measurements, two sensors will always show the same correct value, while the third will present a divergent reading — this is the faulty sensor that needs to be identified quickly.

Help Marina identify which value is incorrect so the team can adjust the manufacturing process immediately. The company's reputation as a leader in reliable sensor technology is in your hands!

Input

The first line of the input contains an integer ($1 \leq T \leq 10^4$), the number of test cases.

The next T lines contain 3 integers, ($-10^9 \leq K_1, K_2, K_3 \leq 10^9$), the values read from each sensor.

It is guaranteed that exactly two of these values will be equal.

Output

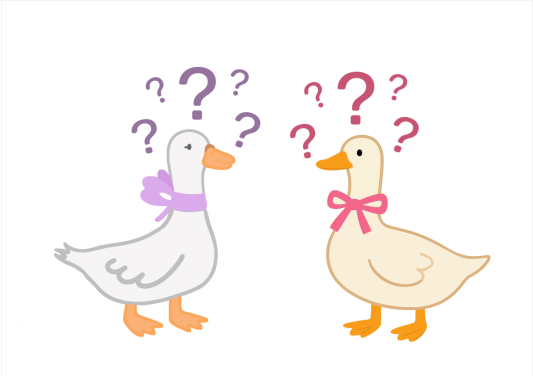
Print a single integer: the value of the sensor that shows a reading different from the other two.

Example

standard input	standard output
5 7 10 7 2 1 1 100 59 59 -42 -42 13 21 3 3	10 2 100 13 21

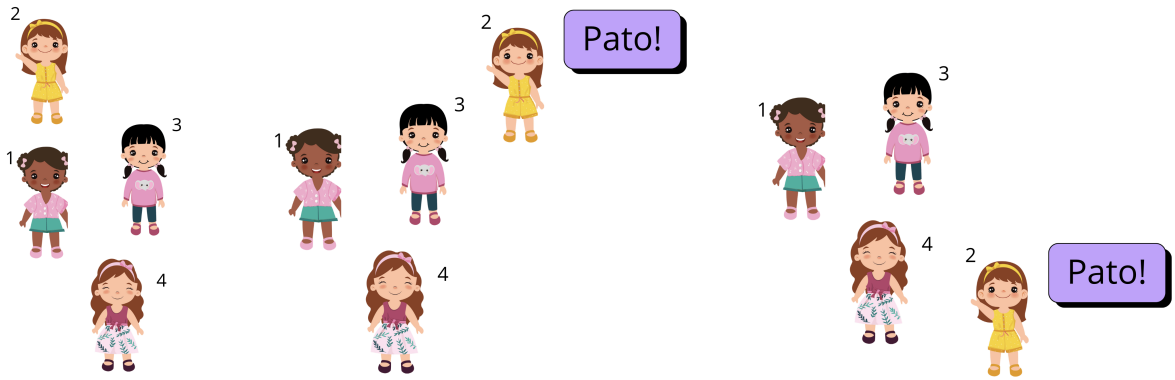
Duck Duck Goose

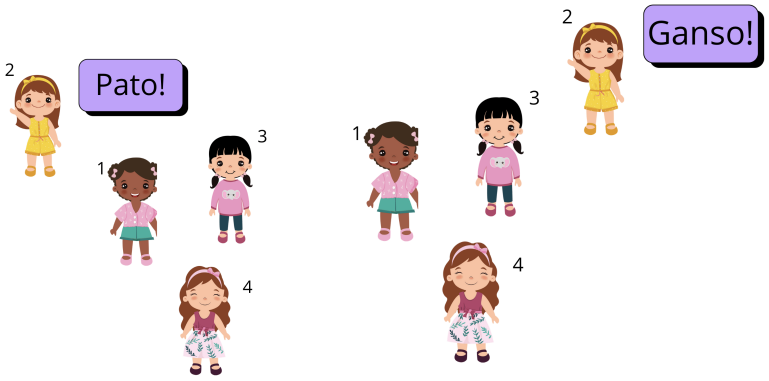
Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes



During school recess, some children gather in a circle to play "Duck-duck-goose." The game works as follows:

- In the circle, there are N children, identified by the integers from 1 to N, such that child i is next to child $i+1$ for $i < N$, and child 1 is next to child N. For example, child 1 is next to children N and 2, child 2 is next to 1 and 3, and so on.
- Initially, all the children are seated, facing the center.
- Through a game of "Eeny, meeny, miny, moe," the children choose a child P, who gets up from the circle. That child begins walking around the circle and says "Duck" while lightly tapping each child's head.
- Finally, after saying "Duck" K times, they shout "Goose" and tag the next child, who must stand up and chase child number P.





Example of a game with $N = 4$ children, where child $P = 2$ starts outside the circle and says "Duck" $K = 3$ times.

One important detail is that since child number P got up from the circle, only $N - 1$ children remain seated: $1, 2, \cdots, P - 1, P + 1, \cdots, N - 1, N$.

After many recesses playing the game, the children noticed that some were saying "Duck" much more than others, who had to wait disproportionately long while sitting. To address this, they decided that the number K of times a child says "Duck" would be fixed.

However, this created a new problem: the game became predictable. As a passionate member of the Female Duck-Duck-Goose Monitors Club (MFP, in Portuguese), you have been tasked with figuring out exactly which child will hear "Goose."

Input

The input consists of three lines, each containing an integer.

The first line contains the integer N, the number of children in the circle. It is guaranteed that $2 \leq N \leq 1000$.

The second line contains an integer P, the identifier of the child who gets up from the circle. It is guaranteed that $1 \leq P \leq N$.

The third line contains an integer K, representing the number of times "Duck" is said. It is guaranteed that $1 \leq K \leq 1000$.

Output

Print a single line containing the position of the child who hears "Goose."

Examples

standard input	standard output
3 1 3	3
2 2 100	1
4 2 3	3

Strengthening Passwords

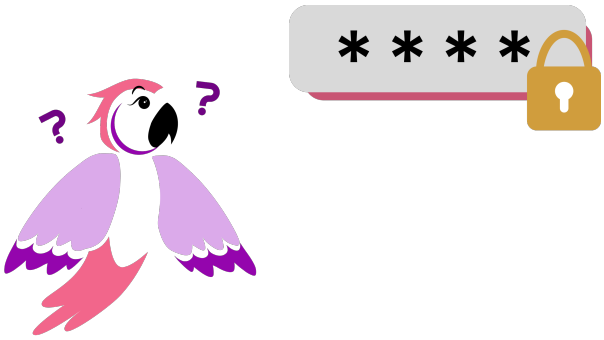
Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Ada is studying programming and has created an amazing website that is very successful! But now she faces a problem, her users create accounts with a variety of passwords, some of which are very weak, for example: 123 and abc.

Concerned about the security of her users, Ada decided to implement a password checker and wants it to be as efficient as possible, so that it only accepts strong passwords.

A password is considered strong by Ada if:

1. it contains more than 10 characters;
2. it has letters and digits;
3. no digit appears after a letter;
4. all digits are ordered in non-decreasing order;
5. all letters are ordered in non-decreasing order.



Help Ada by implementing the checker that determines whether a password is strong!

Input

The first line of input contains an integer N ($1 \leq N \leq 10^5$), which represents the number of passwords.

Each of the next N lines contains a string S ($1 \leq |S| \leq 100$), which consists only of lowercase Latin letters or digits.

Output

For each test case, print "S" if the password is considered strong by Ada and "N" otherwise.

Example

standard input	standard output
5 0adaadaada 123456789abcdefghi jsiwksk2939 mfp2025 2005gpf	N S N N N

Note

Non-decreasing order means that when reading the sequence from left to right, no element is less than its predecessor. In other words, each subsequent character is greater than or equal to the immediately preceding character.

Ring Game

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

After making up since the first time they met in the forest, Ada, the macaw, decides to invite her new friend Py, the snake, for a game of the latest 2D platformer, "Rings".

Rings is a very fun cooperative game, where each player uses a character from various options, so that each character affects the gameplay differently. The game consists of defeating enemies, collecting rings, and earning points!

Ada and Py have chosen their favorite characters and want to collect as many points as they can:

1. Ada chose a strong pink hedgehog. For this character, each ring collected gives P_{pink} points.
2. Py chose a fast blue hedgehog. For this character, each ring collected gives P_{blue} points.

The final score of the game is calculated by multiplying the score each character received. For example, if $P_{pink} = 3, P_{blue} = 2$, Ada collects one ring and Py collects two, the individual scores are, respectively, 3 and 4, but the final score of the team is $3 \cdot 4 = 12$.



Ada and Py know that together they can collect A rings, but their final score depends on how many rings each one collected, so they do not know what the highest possible score they can achieve is. Help them find this maximum!

Input

The first line contains two integers $0 \leq P_{pink}, P_{blue} \leq 10^4$ as described in the statement.

The second line contains an integer $0 \leq A \leq 10^5$, the total number of rings collected.

Output

Print a single integer P, the highest possible score that Ada and Py can achieve in the game Rings.

Example

standard input	standard output
3 2 3	12